Synthetic Image Detection: A Burning Challenge

Problem Statement

The detection of AI generated images from real ones has recently been a challenge in AI forensics. While State-of-the-Art detectors ranging from Transformer-based architectures to Hybrid Spectral-Semantic models achieve near perfect accuracy on trained data, they suffer tremendously on generalization capabilities on cross model datasets. Beyond architectural limitations, these models exhibit a fragility to post-processing, where common operations like JPEG compression and resizing erase the microscopic 'fingerprints' (spectral signatures) that detectors rely on.

Major Weaknesses of Existing Models

The failures of SOTA models can be categorized into three distinct "bottlenecks":

1. **Low-Level Signal Dependence**: Many detectors focus on Fourier transform artifacts or pixel-level co-occurrence. While precise, these signals are the first to be destroyed by social media compression (JPEG) or downscaling.
2. **Generator Overfitting**: Models often overfit to the specific "noise" of a generator's upsampling layer. A detector trained on GANs (like StyleGAN) often has near-zero transferability to Diffusion models (like Midjourney or DALL-E 3).
3. **Calibration and Threshold Bias**: Even if a model identifies a "fake" feature, its internal confidence threshold is often tuned to a specific training set. In "in the wild" scenarios, this leads to a high False Negative Rate (FNR).

Comparison of Robustness

To better understand how these models fail, consider the performance degradation under standard "Social Media" conditions:

- Clean Images: SOTA models often reach 99%+ Accuracy.
- Resized (0.5x): Accuracy typically drops to 70–75%.
- Compressed (JPEG Quality 75): Accuracy often collapses to 60%, which is barely better than a coin flip.

Objective

The objective is to design and evaluate an architecture that is robust to image post processing like image compression and resizing and also that generalizes well in new generation engines.

Methodology

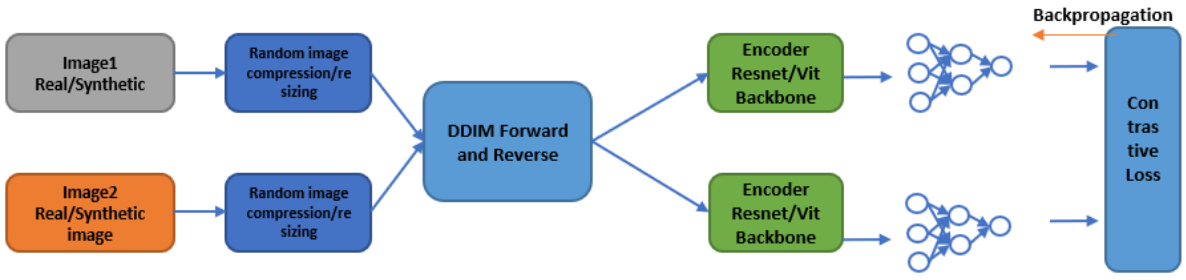


Fig: Architecture Pipeline

1. **Data Input and Pre-Processing**
   The goal is robustness(invariance). The model should learn to ignore some pixel level alterations and focus on extracting invariant features. So, two images selected from two different domain set A (real image) and set B (Synthetic image) are subjected to image preprocessing which consists of Normalization and random image compression and resizing. The selected images are either both real or both synthetic or one real and one synthetic in different iterations.
2. **DDIM forward and reverse process**
   The assumption is that when images are passed to DDIM trained on real images, the denoising process may behave differently to the real and synthetic images. At a specific timestep t, the images are noised and denoised. This is done at random timestep t in each iteration. The output of DDIM is a pair of partially denoised images which are then passed to the encoder. In a way we are performing preprocessing for important features extraction.

   Key hypothesis:
   Manifold Sensitivity: Synthetic images may reside on a slightly different manifold than real images (due to GAN artifacts or Diffusion sampling patterns).
   The "Reconstruction Gap": When a real image is noised and denoised by a model trained on a specific dataset, the "reconstruction error" may be different from that of synthetic image that might have been generated by different process like GANs and Diffusion models.
   Feature Amplification: At specific t values, the Diffusion model may amplify the "unnatural" high-frequency components of synthetic images, making them easier for the Contrastive Encoder to catch.
3. **Encoder**

   The transformed image pairs from the DDIM stage are fed through a deep convolutional neural network (e.g., ResNet50) or a Vision Transformer (e.g., ViT-B/16). It is further connected with Multilayer Perceptron to finally give a latent vector for every input image.

4. **Contrastive Loss and Backpropagation**
   This is the training objective that pulls similar examples together and pushes dissimilar examples apart in the latent space. The core idea of "unique representations" is achieved here. We consider the latent vectors of images of same set as Positive pairs and that of different set as negative pair. For positive pairs, the network is forced to learn their latent representations highly similar. Since they originate from different random compression/resizing, this forces the model to ignore those distortions and find latent features that are consistent.
   The Contrastive Loss function calculates the similarity of a positive pair against the sum of similarities of all possible negative pairs involving one of the positive elements. By minimizing this loss, we are forcing the model to map all transformed synthetic image views (even highly distorted ones) to the same local region in the latent space and explicitly map all transformed real image views (regardless of semantic similarity like a face) to a completely disjoint and separate local region in Z.

   We use the Contrastive loss function used in the paper “A Simple Framework for Contrastive Learning of Visual Representations, Chen et.al”.

$$\ell_{i,j} = -\log \frac{\exp(\mathrm{sim}(\boldsymbol{z}_i, \boldsymbol{z}_j)/\tau)}{\sum_{k=1}^{2N} \mathbb{1}_{[k \neq i]} \exp(\mathrm{sim}(\boldsymbol{z}_i, \boldsymbol{z}_k)/\tau)}$$

   Zi and zj are positive pairs. τ denotes temperature parameter.

In conclusion, we are forcing the encoder to learn unique representations for real and synthetic images irrespective of the image transformations. We further leverage the outputs of DDIM such that model finds some unique factors for learning. The result will be an encoder that has learnt unique disjoint representation for two sets. This encoder can then be used with a classifier for classification and detection of real and synthetic images.